6 Read the passage and answer the questions that follow.

The Juno spacecraft is launched by NASA in 2011, and its mission is to study Jupiter's origin, interior structure, atmosphere, and magnetosphere to understand the solar system's formation by measuring composition, gravity, and magnetic fields with its myriad of sensors on board.

This is also NASA's first mission to Jupiter to use solar panels rather than radioisotope thermoelectric generators (RTGs) for powering the spacecraft. Jupiter orbits the Sun at a much greater distance than Earth, making solar power generation a significant engineering challenge.

Solar radiation spreads out as it travels from the Sun. The solar intensity I at a distance d from the Sun follows the inverse square law. The solar intensity on Jupiter is only 3.72 % that on Earth. To compensate for the low intensity at Jupiter, Juno is equipped with three large solar array wings, as shown in Fig 6.1.

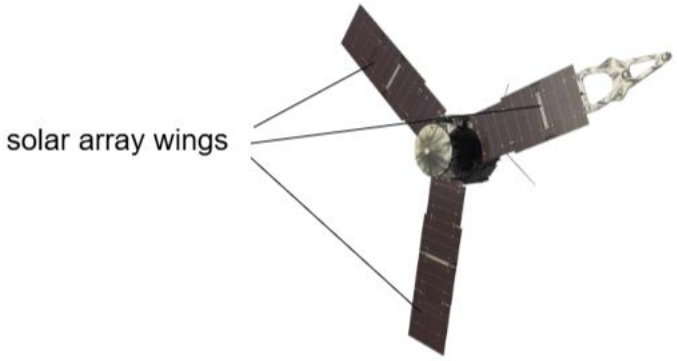


Fig. 6.1

The electrical power P generated by the solar panels on the wings is given by the equation:

$$P = I A \eta$$

where A is the total surface area of the solar panels and η is the efficiency of the solar cells. The efficiency of the cells is affected by the extremely low temperatures in the outer solar system.

Jupiter has an intense radiation belt above its equator that spans a distance of $2R_J$ to $10R_J$ from its centre, where R_J is the radius of Jupiter. Exposure to excessive radiation can cause damage to the spacecraft and its sensors. To minimize such exposure, Juno follows a highly elliptical polar (pole-to-pole) orbit, as shown in Fig 6.2.

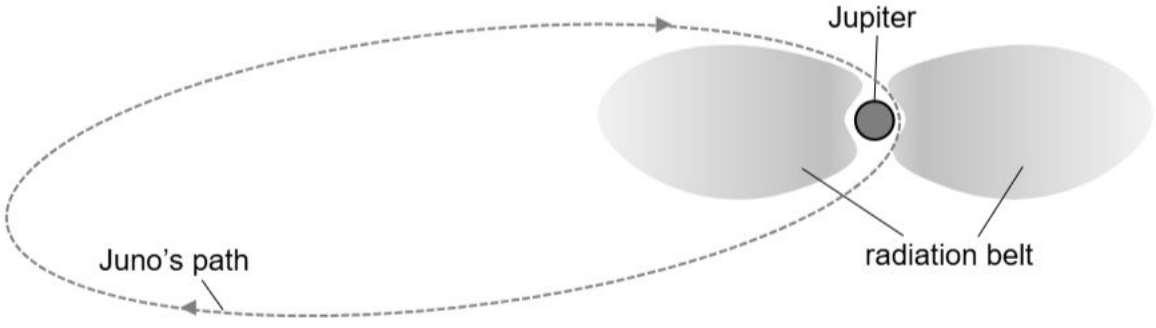


Fig. 6.2

Once Juno enters Jupiter's orbit, its propulsion system is turned off. It is subsequently used sparingly only to correct its path when necessary.

The distance of Juno from the centre of Jupiter in the first three months of 2017 is shown in Fig. 6.3.

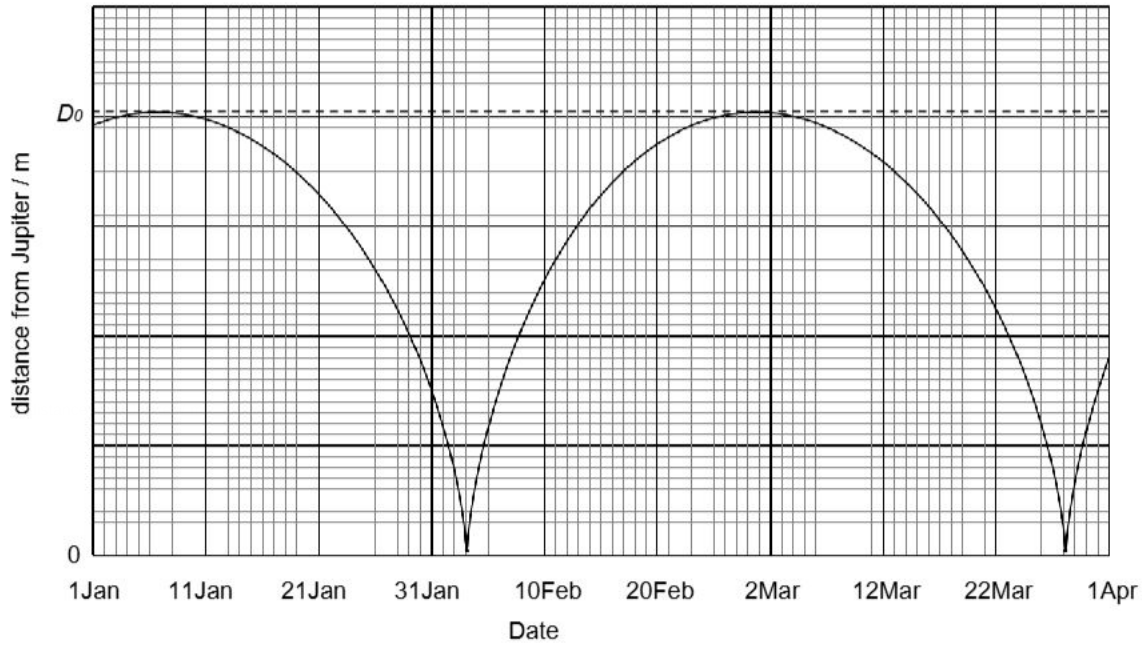


Fig. 6.3

Due to the vast distances involved, it took almost 5 years for Juno to enter Jupiter's orbit from Earth. Communication between Juno and Earth which is conducted via radio waves also experiences significant "light time" delay for signals.

Additional data for Earth, Jupiter and the Juno mission are provided in Table 6.4.

Property	**Value**
Mass of Jupiter M_J	1.90×10^{27} kg
Radius of Jupiter R_J	7.15×10^{7} m
Mean distance* of Earth from the Sun d_E	1.50×10^{11} m
Solar intensity at Earth's distance I_E	1360 W m^{-2}
Total surface area of Juno's solar panels A	60 m^2
Efficiency of solar cells at Jupiter η	24%
Mass of Juno spacecraft m	3625 kg
Perijove (closest distance* to Jupiter) r_p	1.06 R_J
Apojove (furthest distance* from Jupiter) r_a	113 R_J

*center to center

Table 6.4

(a) Explain why the solar intensity I follows an inverse square law as the distance d from the Sun increases.

Sun is taken as a point source, and the light radiates uniformly radially spherical surface area is proportional to the square of distance from the Sun [B1]. Since intensity is inversely proportional to area [B1], it is inversely proportional to the square of the distance. [2]

(b) Using the relation in **(a)** and the data provided, show that the mean distance of Jupiter from the Sun is 7.78×10^{11} m.

$$\frac{\text{Intensity on Jupiter}}{\text{Intensity on Earth}} = 0.0372 = \left(\frac{\text{Distance of Earth}}{\text{Distance of Jupiter}}\right)^2 \quad \text{[B1]}$$

$$\text{Distance of Jupiter} = \frac{1.50\times10^{11}}{\sqrt{0.0372}}\ \text{[B1]} = 7.78\times10^{11}\ \text{m}$$

[2]

(c) Calculate the total electrical power output generated by Juno's solar panels when Jupiter is at its mean distance from the Sun.

$P = I\,A\,\eta = 50.6 \times 60 \times 0.24$

$= 729$ W [A1]

electrical power = ... W [1]

(d) Suggest **one** reason, other than distance from the Sun, why the electrical power output might fluctuate during the mission.

Facing direction of the solar wings relative to the Sun / Jupiter positioned between Juno and the Sun / Any other reasonable answer. [A1] [1]

(e) Determine from Fig. 6.3 the period of Juno's elliptical orbit.

86 – 33 = 53 days [A1]

period = ... days [1]

(f) Determine the value of D_0 in Fig. 6.3.

D_0 = apojove = $113 \times 7.15\times10^7 = 8.08\times10^9$ m [A1]

D_0 = ... m [1]

(g) Explain why the sum of Juno's kinetic energy and gravitational potential energy remains constant when it is in orbit.

With the propulsion system turned off, the only force that acts on Juno is the gravitational force by Jupiter. Since there are no external forces [B1] on the system of Juno and Jupiter, total energy must remain constant. [1]

(h) Calculate the difference between the maximum and minimum values for the kinetic energy of Juno during an orbit.

Difference in max and min values of KE = Difference in max and min values of GPE [B1]

$$= -GM_j m_j \left(\frac{1}{r_a} - \frac{1}{r_p} \right)$$

$$= -6.67 \times 10^{-11} \times 1.90 \times 10^{27} \times 3625 \times \left(\frac{1}{8.08 \times 10^9} - \frac{1}{1.06 \times 7.15 \times 10^7} \right) \text{ [B1]}$$

$$= 6.00 \times 10^{12} \text{ J [A1]}$$

difference = ... J [3]

(i) On Fig. 6.2, indicate point X at which Juno is moving with maximum speed and point Y at which it is moving with minimum speed. [1]

(j) Explain using Fig. 6.2 and your answer in **(i)** how the highly elliptical orbit minimizes the spacecraft's exposure to Jupiter's intense radiation belt.

Fig. 8.2 shows that Juno is near or within the radiation belt for only a small portion of its orbit. [B1] In addition, from **(i)**, its speed when it is near or within the radiation belt is the greatest, hence the time it spends in the region is further minimised. [B1]

[2]

(k) For a body in a circular orbit, gravitational field strength g is always equal to $\frac{v^2}{r}$, where v is its linear speed and r is the radius of the orbit. In Juno's case, assume g is equal to $\frac{v^2}{r}$ at its perijove and its apojove. Use this relationship to calculate Juno's minimum speed.

Speed is minimum at the apojove:

$$g = \frac{GM_J}{r_a^2} = \frac{6.67 \times 10^{-11} \times 1.90 \times 10^{27}}{\left(8.08 \times 10^9\right)^2} \text{ [B1]} = 1.94 \times 10^{-3} \text{ N kg}^{-1}$$

Since $g = \frac{v^2}{r}$, $v = \sqrt{gr} = \sqrt{1.94 \times 10^{-3} \times 8.08 \times 10^9}$ [B1] $= 3.96 \times 10^3$ m s^{-1} [A1]

minimum speed = ... m s^{-1} [3]

(l) Explain why the relationship $g = \frac{v^2}{r}$ does not hold in general for elliptical orbits.

Juno's distance from Jupiter is not constant. [B1]

When Juno is moving towards Jupiter, $g > v^2/r$. When Juno is moving away from Jupiter, $g < v^2/r$. [B1], OR Juno's velocity v is almost always not perpendicular to g, thus there is a component of v that is causing it to move closer or further away from Jupiter. [B1] [2]

(m) Using the mean distances of Jupiter and of Earth from the Sun, estimate a minimum value for the "light time" delay for signals to travel from Jupiter to Earth.

Minimum distance between Jupiter and Earth = $(7.78 - 1.50)\times10^{11} = 6.28\times10^{11}$ m [B1]

Minimum time taken = $\frac{6.28\times10^{11}}{3.0\times10^{8}}$ = 2100 s [A1]

delay = ... s [2]

[Total: 22]